

SURYAKIRAN SURESHKUMAR

+1 551-222-9708 • suryakiranbds@gmail.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

EXPERIENCE

- Analyst, Software Engineering | **Goldman Sachs** September 2024 - Present
 - Designed and developed a scalable spark refiner that periodically ingests, normalizes, and stores margining and netting terms in a **centralized data lake**, processing 3M+ records per day to support downstream margin calculation pipelines.
 - Built an **MCP-based support assistant** that, given an L3 support query, automatically queries **Snowflake** tables and investigates logs via BQL to surface root causes, **cutting average triage time ~80% (from ~30 min to under 5 per query)** and saving the on-call team significant manual effort.
 - Primary developer for an **LLM-powered system** that automates netting rule creation from structured playbooks, using **MCP-based tool orchestration** to connect the model to internal data sources and rule-emission services, gated by an **evaluation layer** that scores each output and escalates only exceptions to humans, cutting rule creation from weeks to minutes (\$1.2M saved annually).
 - Designed and implemented a **risk analysis platform** to automate trade margining based on legal agreements, enabling margin call workflows and supporting **16M+ daily requests with 99.99% uptime** using Java, Spring Boot, and Kubernetes.
 - Boosted system performance by 30-40% through RabbitMQ-based async processing, HAProxy load balancing, and Apache Geode caching, backed by MySQL for data integrity.
 - Contributed across the full SDLC, from requirements to global deployment, collaborating with **cross-functional teams** across 5+ regions.
- Data Engineer Intern | **Promantus Inc.** June 2023 - August 2023
 - Implemented and **fine-tuned** a few-shot object-detection model for P&ID diagrams, a 10% accuracy improvement over the prior solution.
- Analyst, Machine Learning Engineering | **Tiger Analytics** February 2021 - July 2022
 - As lead engineer, designed and deployed 30+ predefined data-science functions on Azure Databricks and PySpark into an **internal platform engineers relied on daily**, built a scalable **FastAPI** backend, and spearheaded a **CI/CD pipeline** that improved deployment speeds by 20%, collaborating cross-functionally to ship the platform.
 - Led creation of a mobile app for inventory video analysis, slashing manual labor by 60%, and built an **AWS ETL data pipeline** to analyze financial trends and drive data-informed decisions.
- Summer Intern | **National University of Singapore & Hewlett Packard Enterprise** June 2019
 - Built an award-winning Django app using **neural networks** to detect phishing sites, ranked #1 of 40 projects; also implemented AES encryption via a Hadoop MapReduce job.

EDUCATION

- Courant Institute of Mathematical Sciences, New York University** New York, NY
M.S - Computer Science; GPA: 3.85/4.00 *September 2022 - May 2024*
- Thiagarajar College of Engineering, Anna University** India
B.E - Computer Science and Engineering; GPA: 9.11/10.00 *August 2017 - April 2021*

SKILLS SUMMARY

- Programming Languages:** Python, Java, C++, C
- AI / ML:** PyTorch, TensorFlow, LangChain, OpenAI, LLM systems, RAG, MCP, LoRA fine-tuning
- Full-Stack Development:** Spring Boot, HTML, CSS, JavaScript, Django, FastAPI, Node.js, React.js, Flask
- Cloud, Version Control & CI/CD:** AWS, Azure, Oracle, RabbitMQ, Terraform, Git, GitHub, GitLab, Azure DevOps
- Big Data Technologies:** Hadoop, HDFS, YARN, MapReduce, Hive, Elasticsearch, Spark
- Databases:** MySQL, SQL Server, PostgreSQL, MongoDB, Oracle, Redshift, Snowflake, Redis, BigQuery
- Tools & Platforms:** Postman, Power Apps, DataBricks, Docker, Flutter, JIRA

PROJECTS

- DriveVLM:** Expertly fine-tuned(using Low-Rank Adaptation) the QwenVL Vision-Language Model within a Carla Simulator environment for autonomous driving applications, achieving superior performance compared to traditional methods while utilizing only 40% of the data.
- ChatLoom:** Implemented a specialized chatbot using OpenAI's advanced large language model for detailed cosmology and astrophysics responses. Integrated with LangChain, it's accessible via a web application developed using Chainlit.
- Phishing Site Detection:** Developed a web application, complemented by a Chrome extension, that utilizes a neural network to detect phishing sites after rigorous data cleaning, achieving an accuracy of 98.73%. The system notifies users about the safety of sites and forwards analysis to the cyber department.